



Database Testing with Selenium 4 (Java)

(Automation QA/SDET)

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@KushalParikh

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1) Setup SQL Database in your local

1. Download link - <https://dev.mysql.com/downloads/installer/>
2. Click on "Download "

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MySQL Installer

General Availability (GA) Releases Archives ⓘ

MySQL Installer 8.0.41

Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:
8.0.41

Select Operating System:
Microsoft Windows

Windows (x86, 32-bit), MSI Installer (mysql-installer-web-community-8.0.41.0.msi)	8.0.41	2.1M	Download
Windows (x86, 32-bit), MSI Installer (mysql-installer-community-8.0.41.0.msi)	8.0.41	352.2M	Download

MD5: 22ed92c892160254fbf0f93d811360c2 | Signature

MD5: c2e89b80cf89c2214e5ecb9f91b77f10 | Signature

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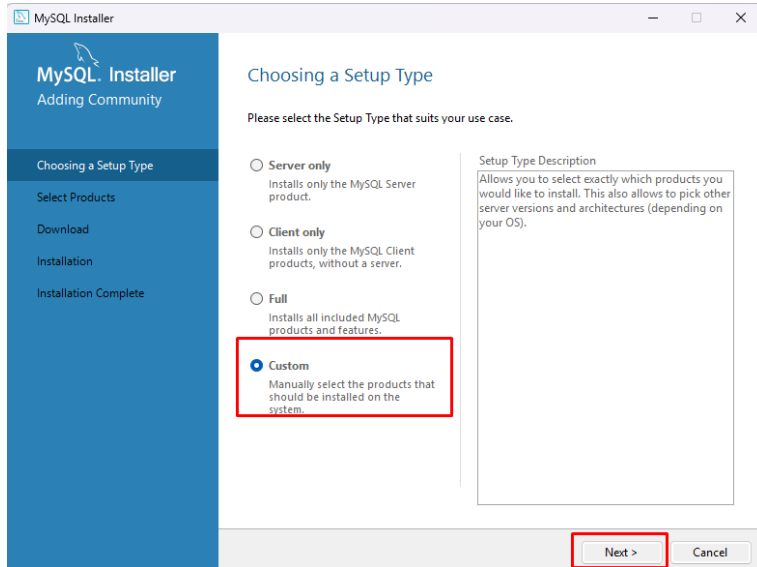
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Wait till download is completed....

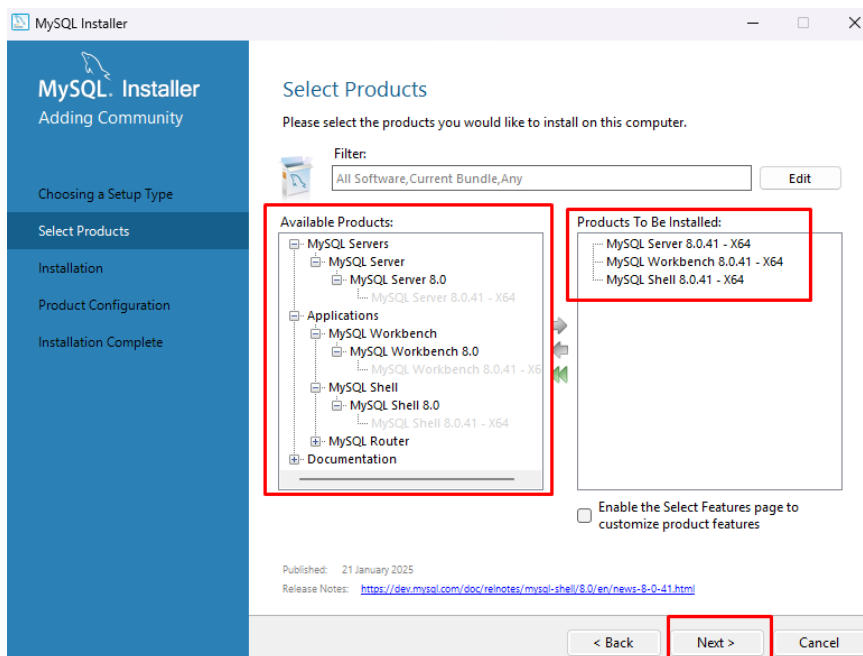
4. Once download is done, click on downloaded installer & choose the “Custom” & click next button

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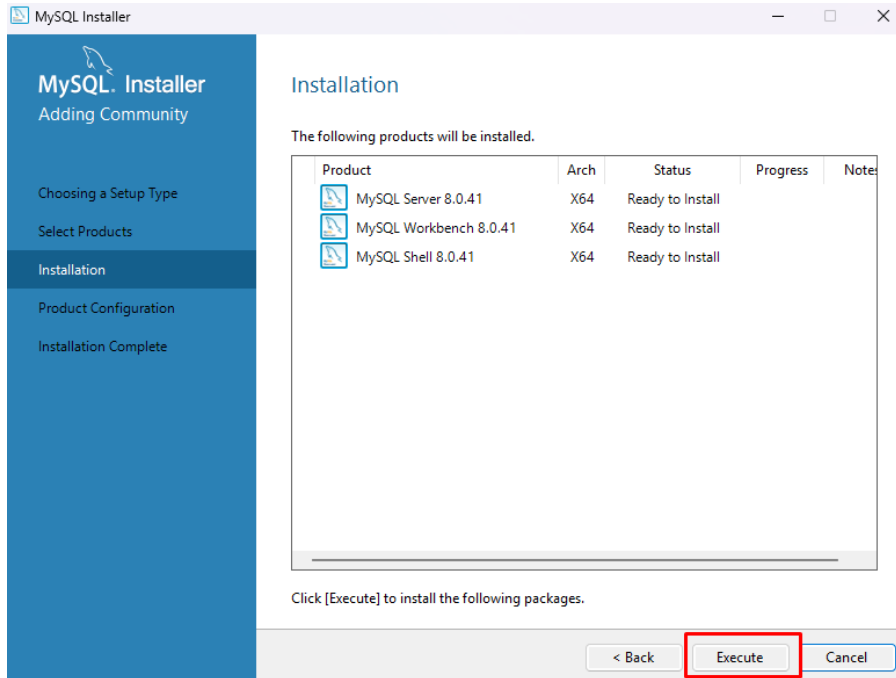


5. Now select the highlighted products & click on next button & then click on Execute button



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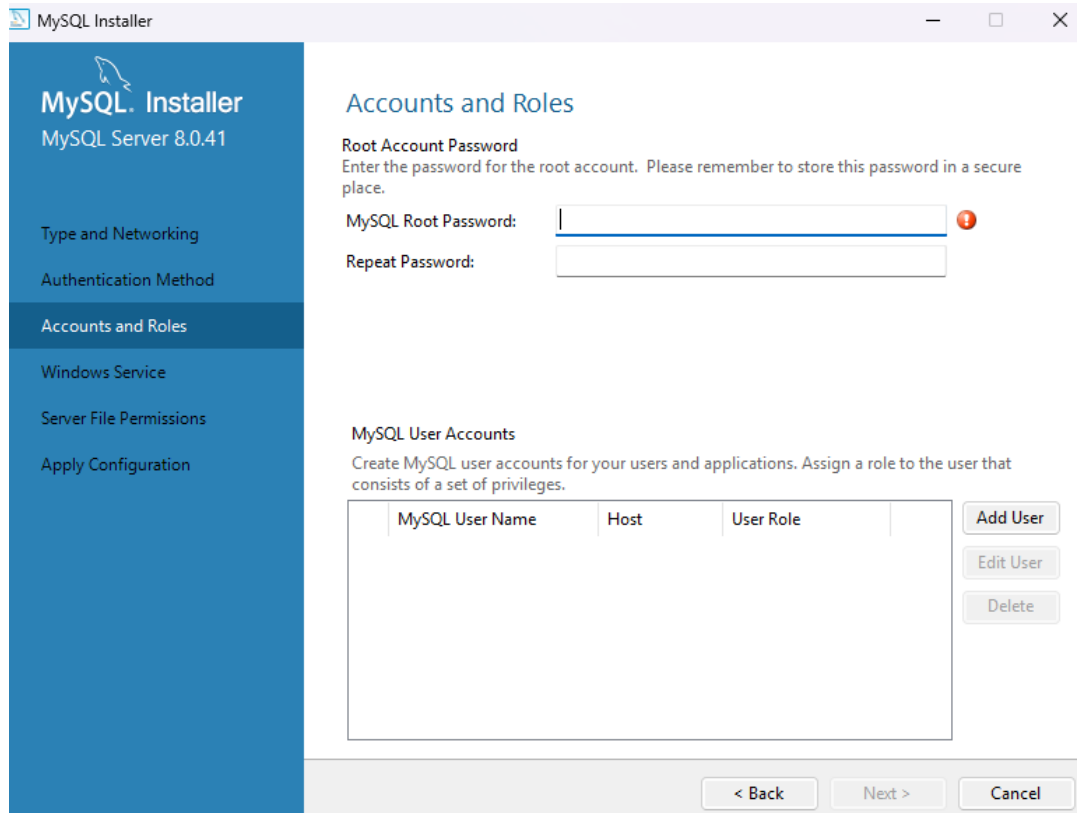


6. Click on next buttons until below screen comes :

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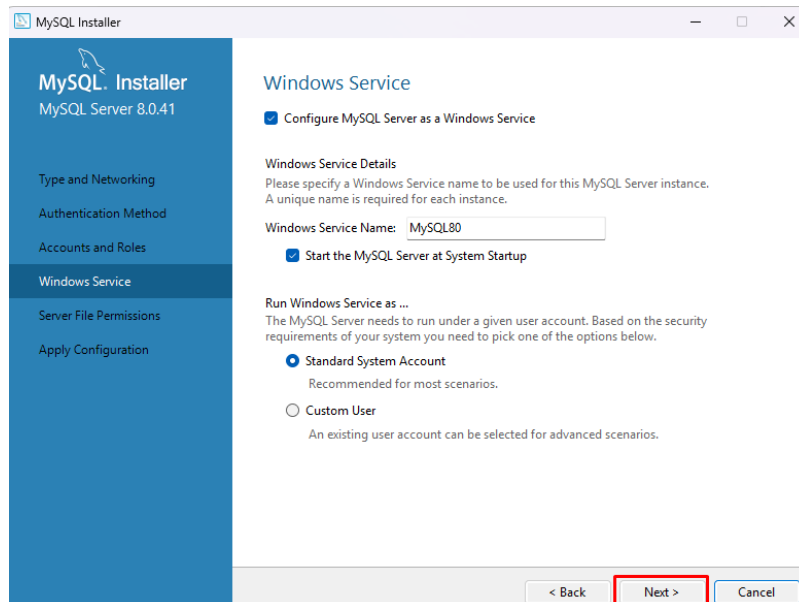
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The screenshot shows the 'Accounts and Roles' step of the MySQL Installer for MySQL Server 8.0.41. The left sidebar lists the installation steps: Type and Networking, Authentication Method, Accounts and Roles (selected), Windows Service, Server File Permissions, and Apply Configuration. The main area is titled 'Accounts and Roles' and contains two sections. The first section, 'Root Account Password', prompts the user to enter a password for the root account, with a note to store it securely. It includes input fields for 'MySQL Root Password' and 'Repeat Password'. The second section, 'MySQL User Accounts', instructs the user to create MySQL user accounts and assign roles. It features a table with columns for 'MySQL User Name', 'Host', and 'User Role'. To the right of the table are buttons for 'Add User', 'Edit User', and 'Delete'. At the bottom of the window are navigation buttons: '< Back', 'Next >', and 'Cancel'.

7. Type your password & click on next button

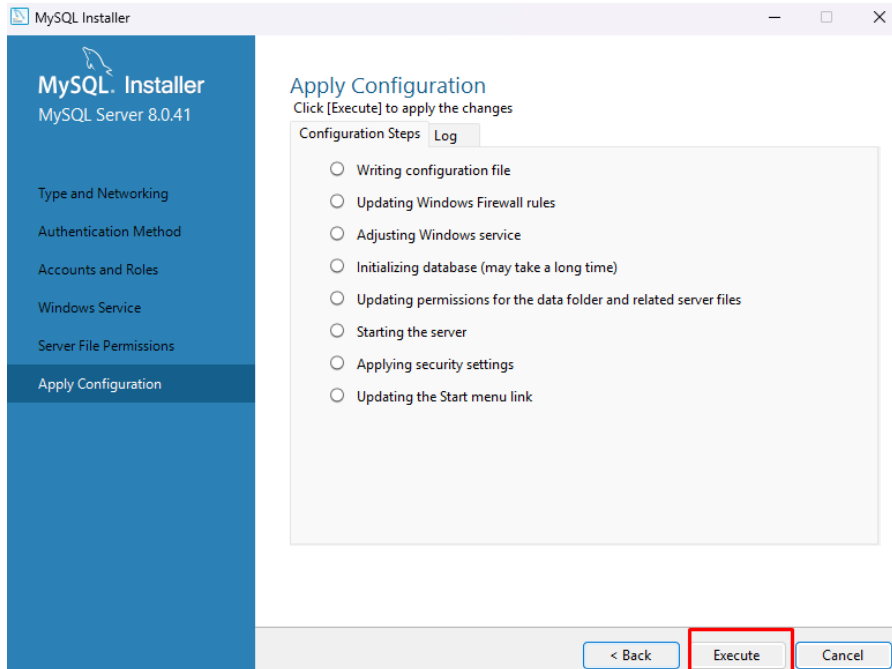


The screenshot shows the 'Windows Service' step of the MySQL Installer for MySQL Server 8.0.41. The left sidebar is the same as in the previous screenshot, with 'Windows Service' now selected. The main area is titled 'Windows Service' and contains several options. A checkbox labeled 'Configure MySQL Server as a Windows Service' is checked. Below this, the 'Windows Service Details' section asks the user to specify a Windows Service name, with a text input field containing 'MySQL80'. Another checkbox, 'Start the MySQL Server at System Startup', is also checked. The 'Run Windows Service as ...' section provides instructions on the user account to use, with two radio button options: 'Standard System Account' (selected) and 'Custom User'. At the bottom, the navigation buttons are '< Back', 'Next >', and 'Cancel'. The 'Next >' button is highlighted with a red rectangle.

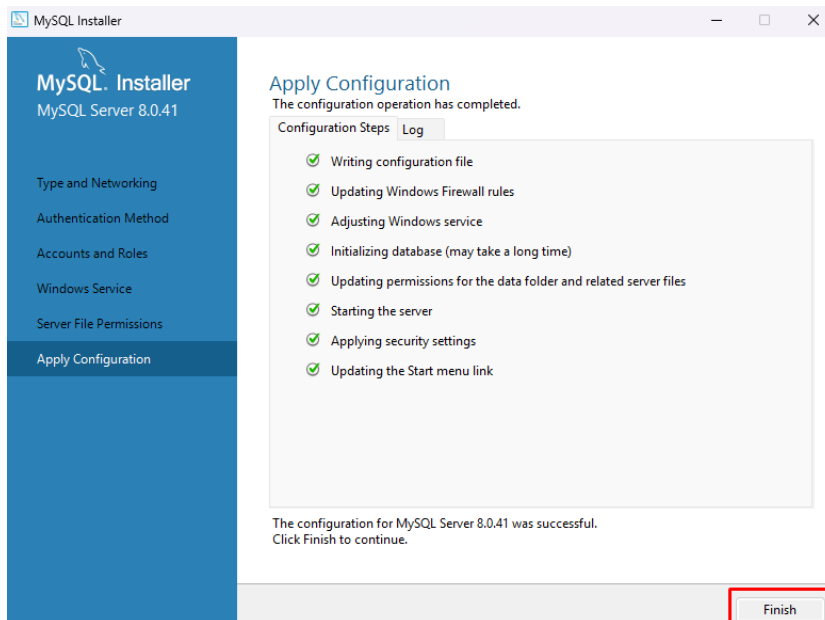
8. Click on Execute button

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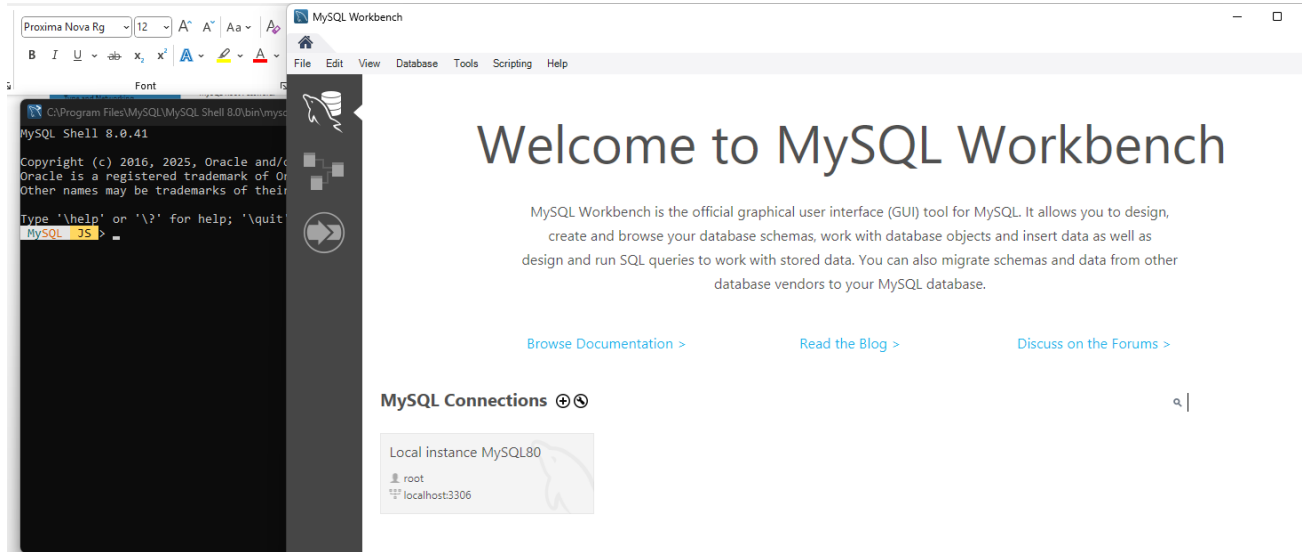
9. Click on Finish button



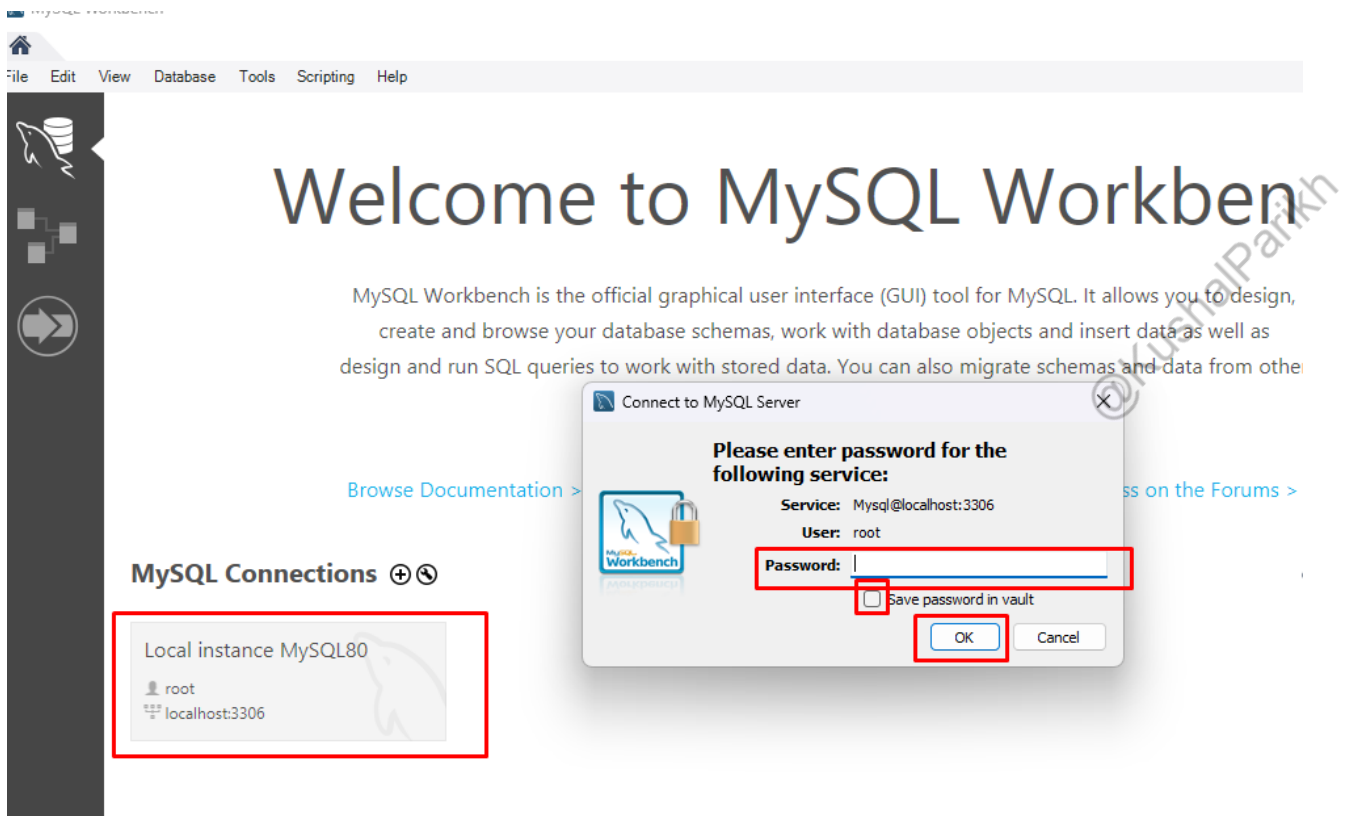
10. Now finish all further steps – click on Finish button & below screen will be open

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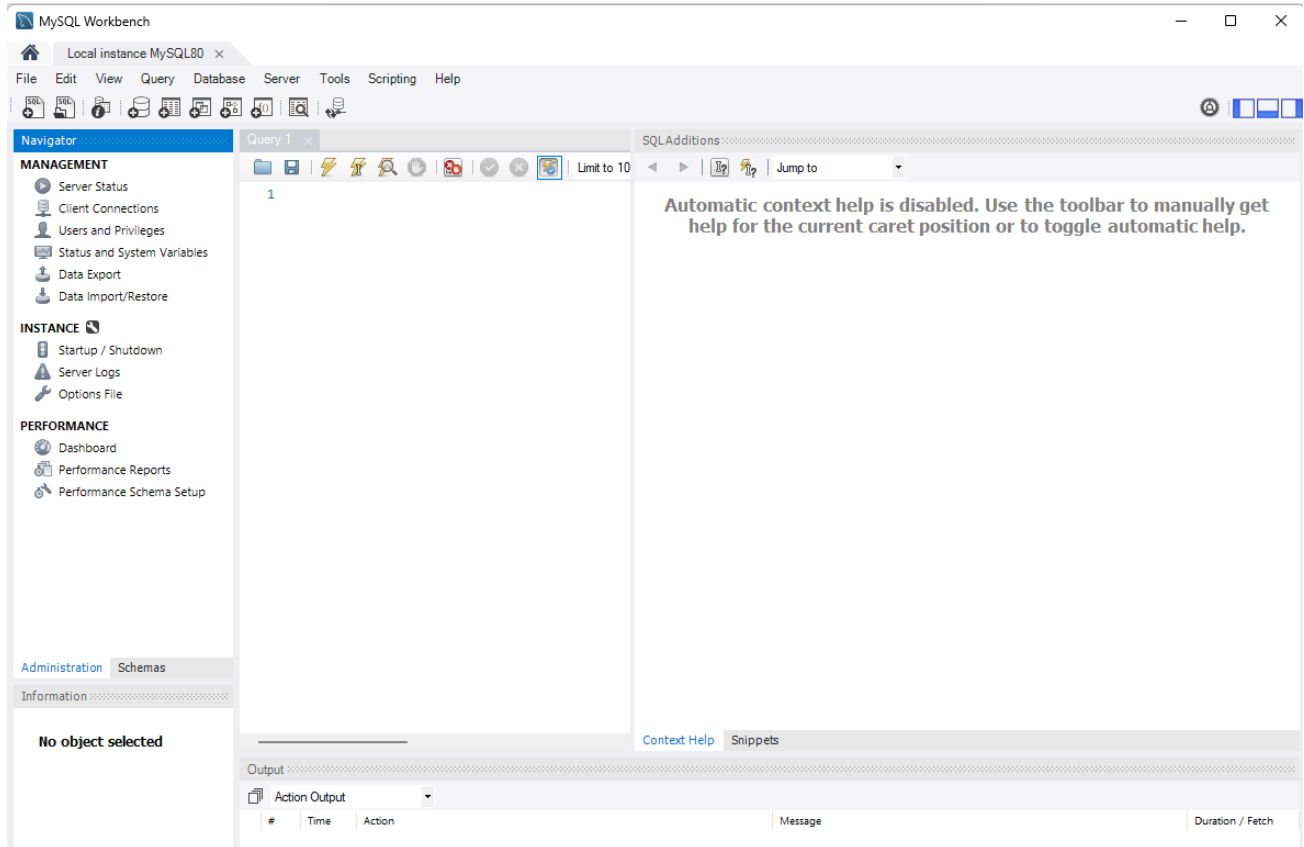
11. Click on highlighted section & enter your password used during setup and click on OK button



12. My SQL Workbench will be open

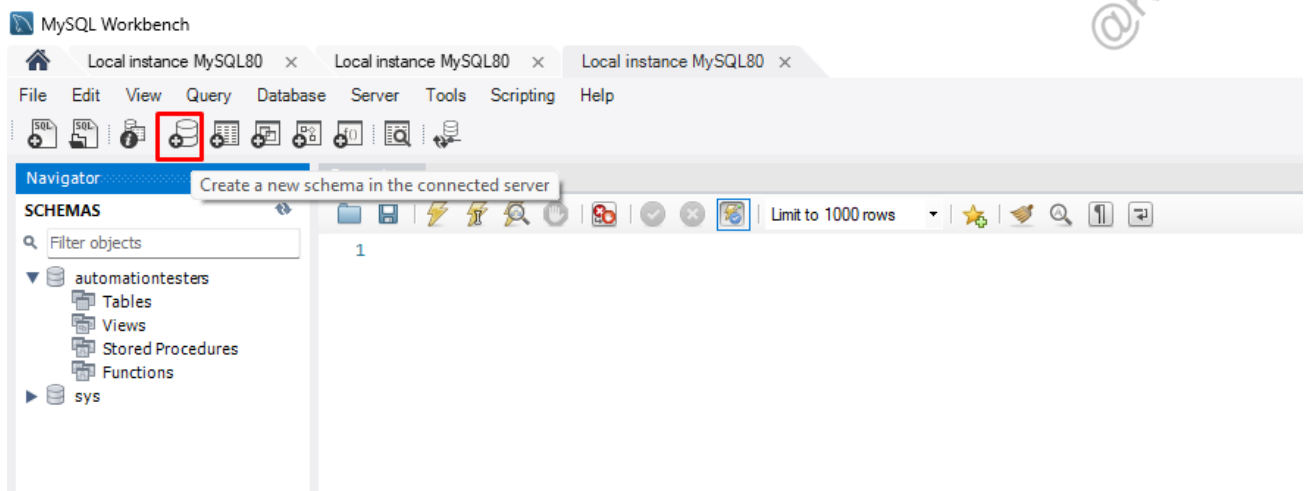
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2) How to create your own SQL Database table for testing purpose ?

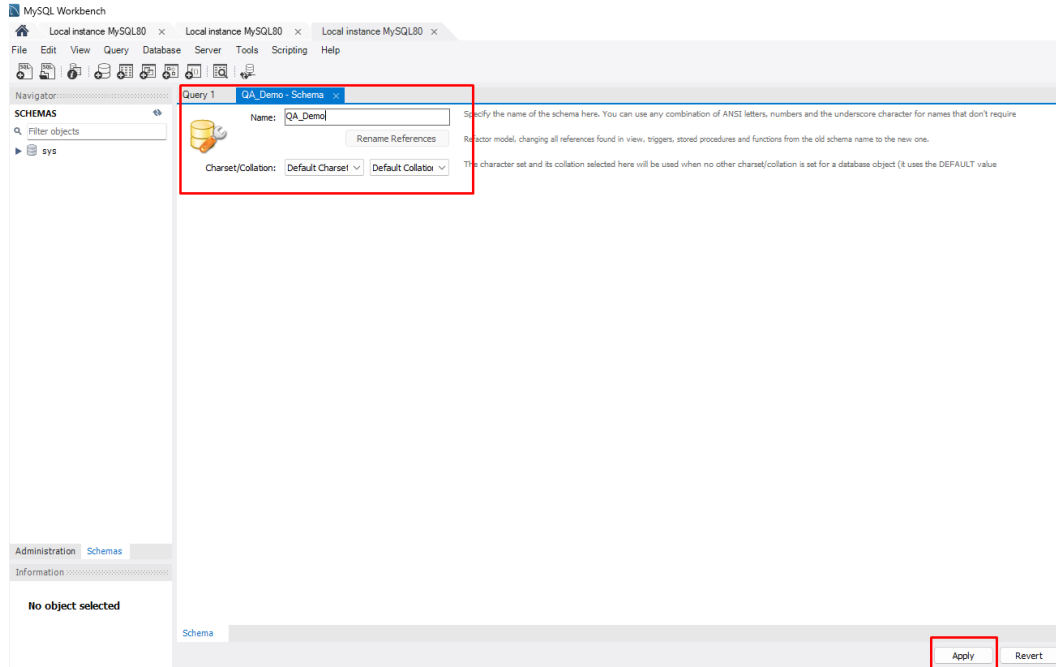
1. Click on "scheme" icon



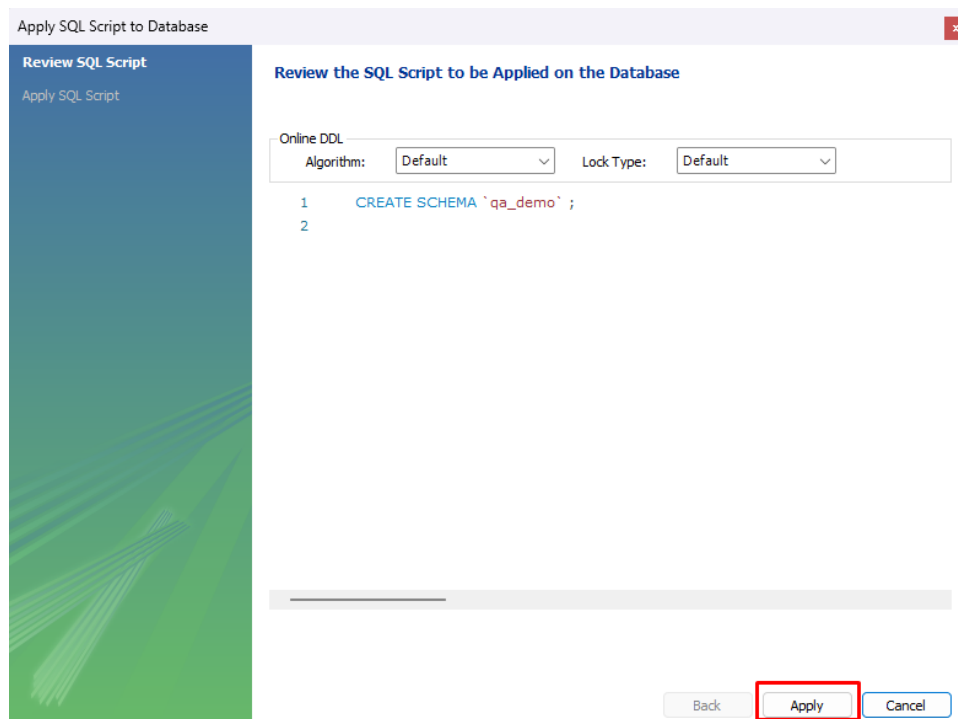
2. Enter your scheme name "Any Name You want" & click on Apply button

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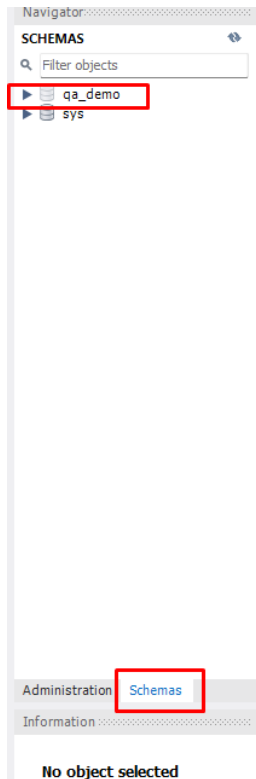


3. Click on Apply button again & click on Finish button

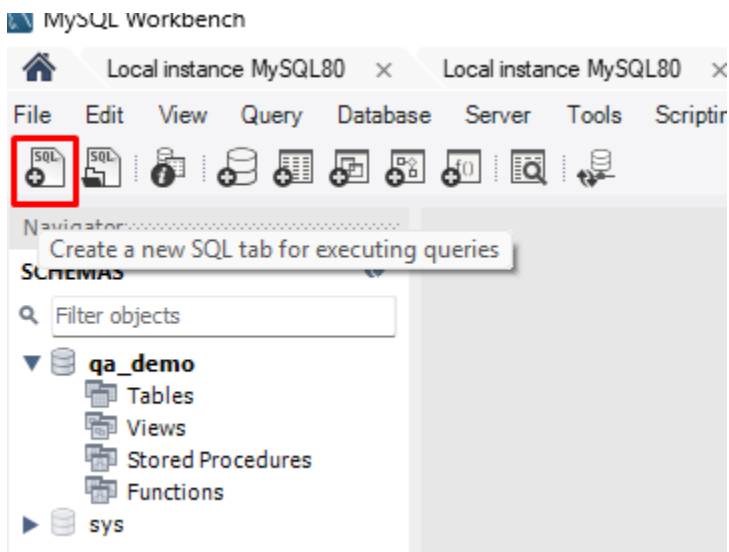


4. Click "Scheme" tab & your scheme is created successfully
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5. Now we will create our records for testing purpose – click on SQL Query Tab



6. Copy paste the below code in query section

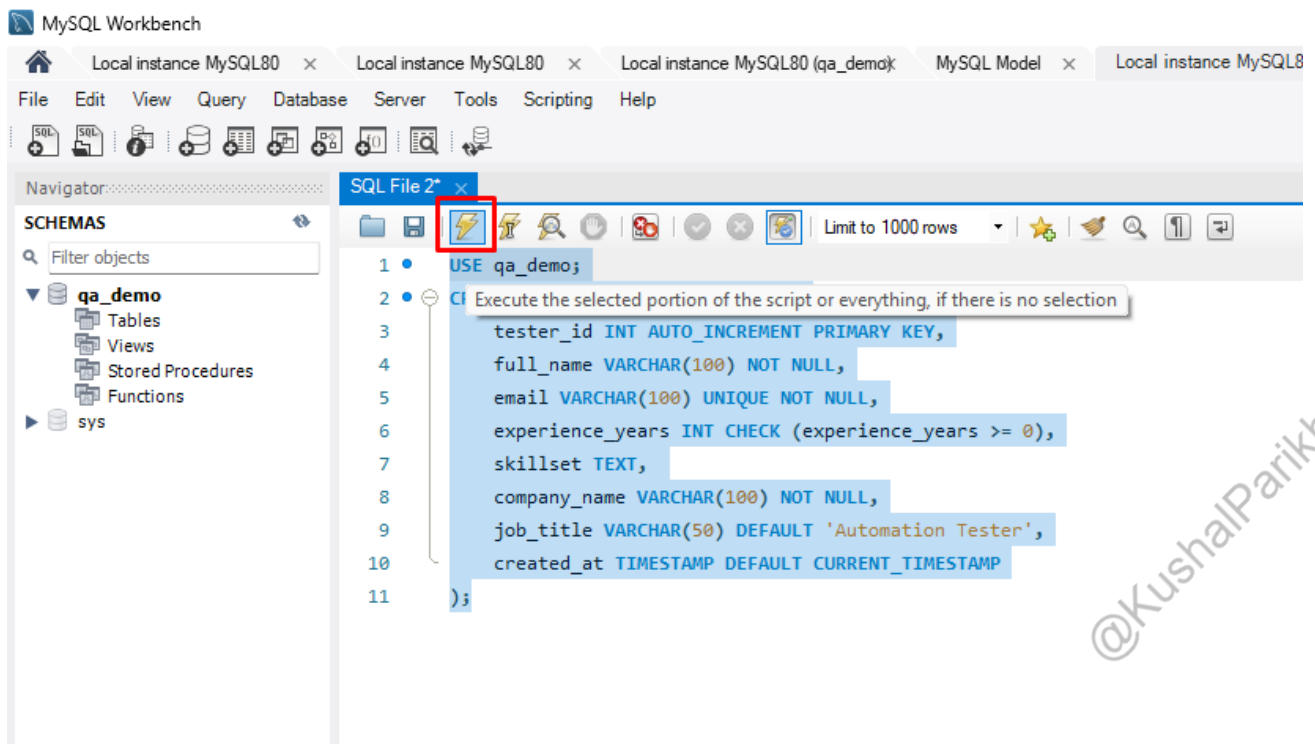
```
USE qa_demo;
```

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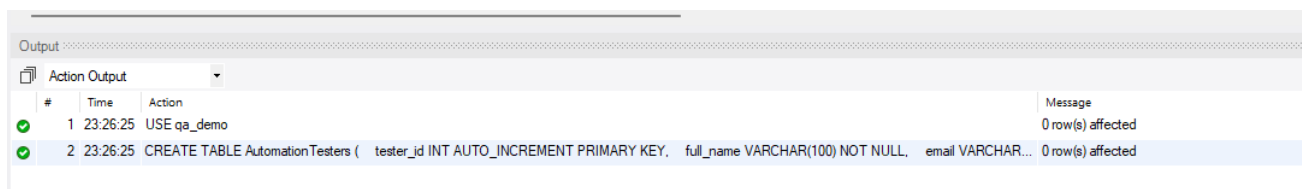
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```
CREATE TABLE AutomationTesters (  
  tester_id INT AUTO_INCREMENT PRIMARY KEY,  
  full_name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) UNIQUE NOT NULL,  
  experience_years INT CHECK (experience_years >= 0),  
  skillset TEXT,  
  company_name VARCHAR(100) NOT NULL,  
  job_title VARCHAR(50) DEFAULT 'Automation Tester',  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

7. Press Ctrl + A & click on "execute" icon



8. Your table has been created successfully



9. Now we will add our test records with SQL queries & then later we will try to add new records using Selenium with JDBC connections

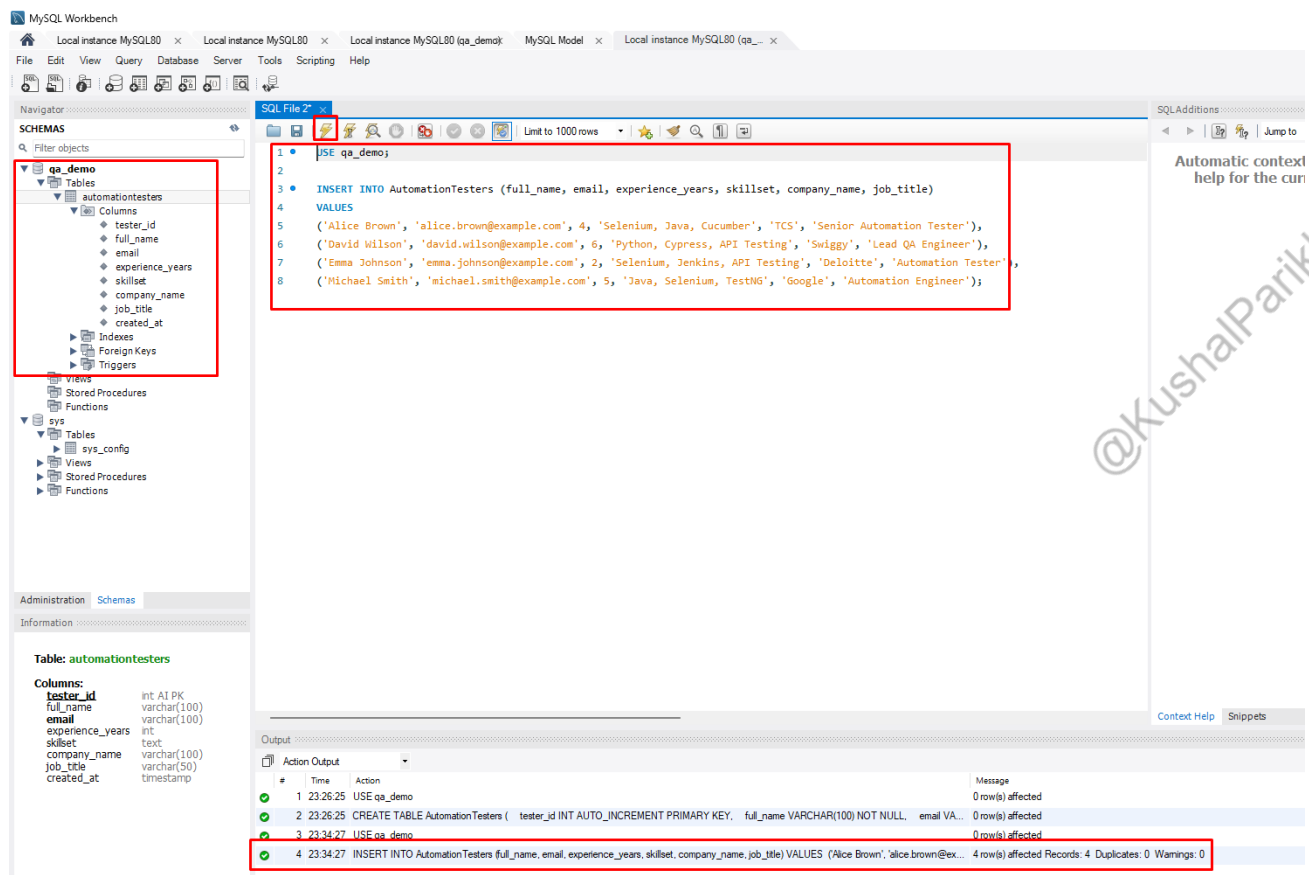
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Use below SQL query to add new records & click on execute icon

```
USE qa_demo;

INSERT INTO AutomationTesters (full_name, email, experience_years, skillset, company_name,
job_title)
VALUES
('Alice Brown', 'alice.brown@example.com', 4, 'Selenium, Java, Cucumber', 'TCS', 'Senior Automation
Tester'),
('David Wilson', 'david.wilson@example.com', 6, 'Python, Cypress, API Testing', 'Swiggy', 'Lead QA
Engineer'),
('Emma Johnson', 'emma.johnson@example.com', 2, 'Selenium, Jenkins, API Testing', 'Deloitte',
'Automation Tester'),
('Michael Smith', 'michael.smith@example.com', 5, 'Java, Selenium, TestNG', 'Google', 'Automation
Engineer');
```



The screenshot shows the MySQL Workbench interface. On the left, the 'Schemas' pane shows the 'qa_demo' database selected, with the 'AutomationTesters' table highlighted. The 'SQL Editor' in the center contains the SQL query from the previous block. The 'Output' window at the bottom shows the execution results:

#	Time	Action	Message
1	23:26:25	USE qa_demo	0 row(s) affected
2	23:26:25	CREATE TABLE AutomationTesters (tester_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email VA...	0 row(s) affected
3	23:34:27	USE qa_demo	0 row(s) affected
4	23:34:27	INSERT INTO AutomationTesters (full_name, email, experience_years, skillset, company_name, job_title) VALUES ('Alice Brown', 'alice.brown@ex...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0

Your records are created successfully!!

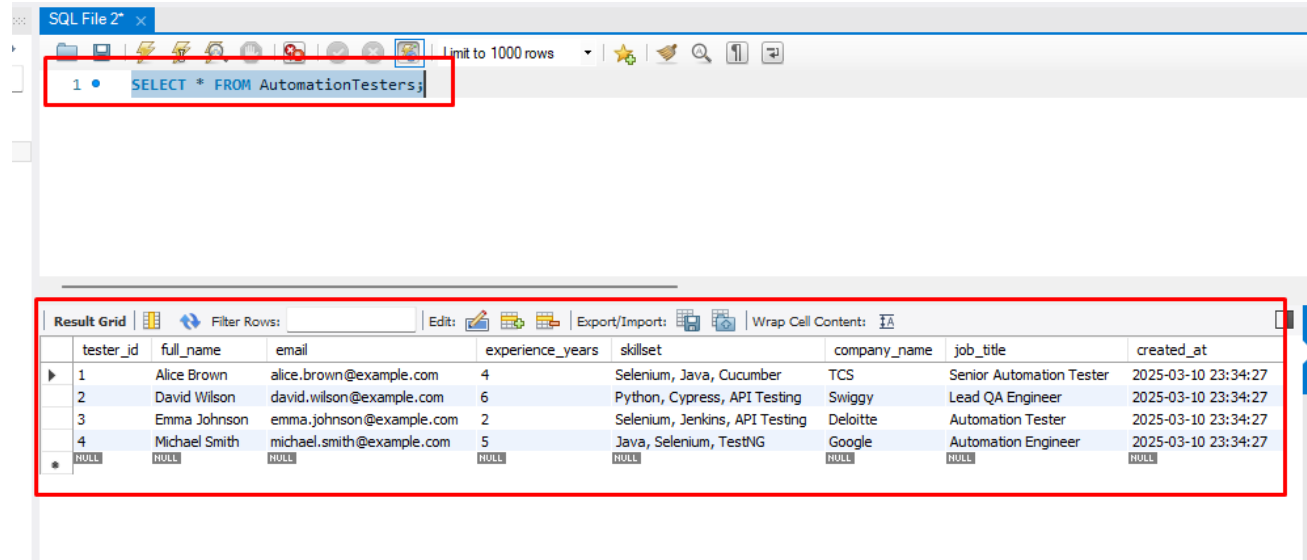
3) How to fetch your SQL Data records ?

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Using Select Query as below :

```
SELECT * FROM AutomationTesters;
```



The screenshot shows a SQL client window titled 'SQL File 2*'. The query editor contains the query 'SELECT * FROM AutomationTesters;'. Below the editor, the 'Result Grid' displays the results of the query. The grid has columns for tester_id, full_name, email, experience_years, skillset, company_name, job_title, and created_at. The results are as follows:

tester_id	full_name	email	experience_years	skillset	company_name	job_title	created_at
1	Alice Brown	alice.brown@example.com	4	Selenium, Java, Cucumber	TCS	Senior Automation Tester	2025-03-10 23:34:27
2	David Wilson	david.wilson@example.com	6	Python, Cypress, API Testing	Swiggy	Lead QA Engineer	2025-03-10 23:34:27
3	Emma Johnson	emma.johnson@example.com	2	Selenium, Jenkins, API Testing	Deloitte	Automation Tester	2025-03-10 23:34:27
4	Michael Smith	michael.smith@example.com	5	Java, Selenium, TestNG	Google	Automation Engineer	2025-03-10 23:34:27
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4) Database testing with Selenium – JDBC Connection

Note:

Selenium is not required for Database testing but what we actually do in real time scenarios . Suppose requirement is –

- 1) we need to create user from frontend and need to verify in database that user is created successfully or not in database**
- 2) Delete user from using delete account functionality & verify in database if user is deleted or not**
- 3) Update user info from using update account functionality & verify in database if user info is updated or not.**

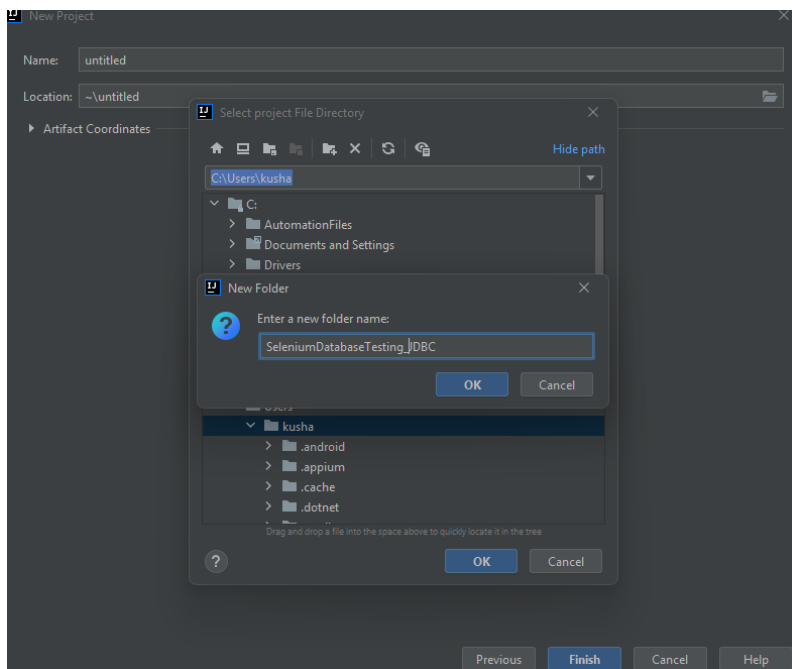
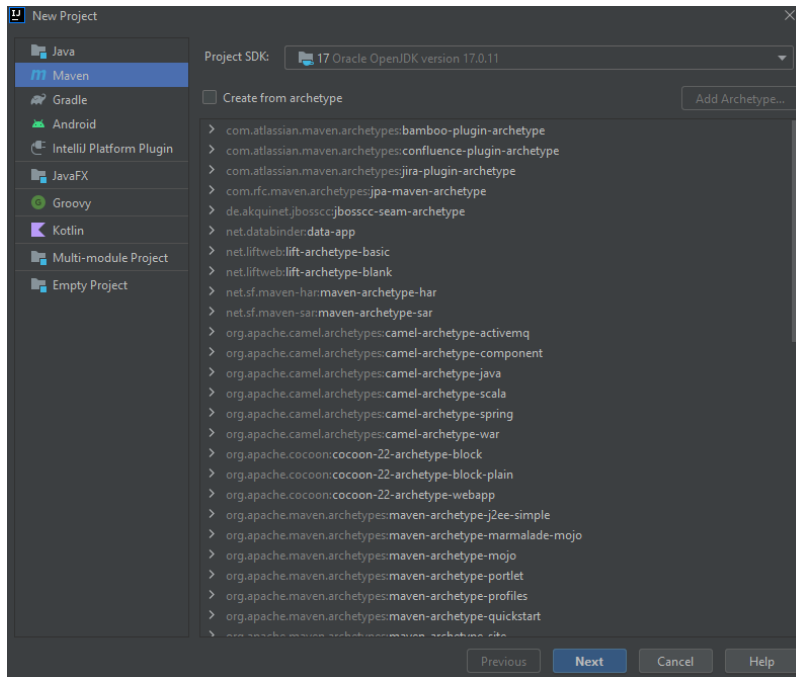
SO DON'T CONFUSE WITH SELENIUM IS REQUIRED FOR DATABASE TESTING

1) Prerequisites Steps :

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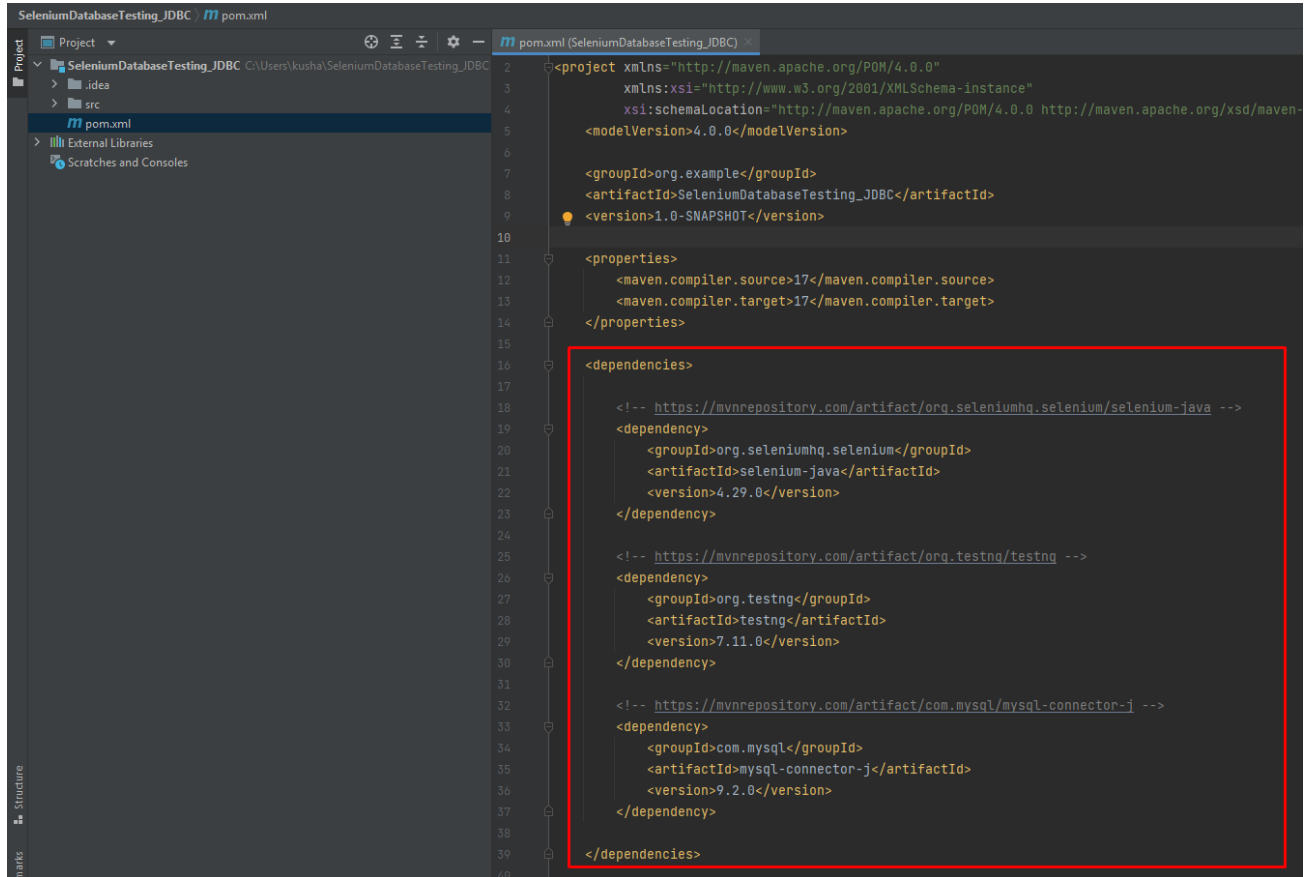
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- Open any IDE , Create Maven Project & add required dependencies as below



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```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0"
3         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-
5         <modelVersion>4.0.0</modelVersion>
6
7         <groupId>org.example</groupId>
8         <artifactId>SeleniumDatabaseTesting_JDBC</artifactId>
9         <version>1.0-SNAPSHOT</version>
10
11         <properties>
12             <maven.compiler.source>17</maven.compiler.source>
13             <maven.compiler.target>17</maven.compiler.target>
14         </properties>
15
16         <dependencies>
17
18             <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
19             <dependency>
20                 <groupId>org.seleniumhq.selenium</groupId>
21                 <artifactId>selenium-java</artifactId>
22                 <version>4.29.0</version>
23             </dependency>
24
25             <!-- https://mvnrepository.com/artifact/org.testng/testng -->
26             <dependency>
27                 <groupId>org.testng</groupId>
28                 <artifactId>testng</artifactId>
29                 <version>7.11.0</version>
30             </dependency>
31
32             <!-- https://mvnrepository.com/artifact/com.mysql/mysql-connector-j -->
33             <dependency>
34                 <groupId>com.mysql</groupId>
35                 <artifactId>mysql-connector-j</artifactId>
36                 <version>9.2.0</version>
37             </dependency>
38
39         </dependencies>
40     </project>
```

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Add below dependencies in POM File :-

<dependencies>

<!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->

<dependency>

<groupId>org.seleniumhq.selenium</groupId>

<artifactId>selenium-java</artifactId>

<version>4.29.0</version>

</dependency>

<!-- https://mvnrepository.com/artifact/org.testng/testng -->

<dependency>

<groupId>org.testng</groupId>

<artifactId>testng</artifactId>

<version>7.11.0</version>

</dependency>

<!-- https://mvnrepository.com/artifact/com.mysql/mysql-connector-j -->

<dependency>

<groupId>com.mysql</groupId>

<artifactId>mysql-connector-j</artifactId>

<version>9.2.0</version>

</dependency>

</dependencies>

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2) Test Scenarios :

- Create a new user (automation tester user) from frontend using Selenium (Java)
- Verify user is created or not by fetching the data from SQL database server

Code :-

```
import org.testng.annotations.Test;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

public class DatabaseTesting {

    @Test    public void CreateUser_Test(){

        System.out.println("Creating User using Selenium from frontend...");

        System.out.println("-----");

        System.out.println("Fetching Data to verify if user is created successfully or not ...");

        // Database connection details
        String url = "jdbc:mysql://localhost:3306/qa_demo"; // Update DB name if different
        String user = "root"; // Change if different
        String password = "Kush@123"; // Change if different

        // SQL Query to fetch all records
        String query = "SELECT * FROM AutomationTesters";

        // Establish Connection
        try (Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery(query)) {

            // Display Column Headers
            System.out.printf("%-5s %-20s %-30s %-5s %-40s %-20s %-25s %-20s%n",
                "ID", "Full Name", "Email", "Exp", "Skillset", "Company", "Job Title", "Created At");
            System.out.println("-----");

            // Iterate over the result set
            while (rs.next()) {
                int id = rs.getInt("tester_id");
                String fullName = rs.getString("full_name");
                String email = rs.getString("email");
```

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```
int experience = rs.getInt("experience_years");
String skillset = rs.getString("skillset");
String company = rs.getString("company_name");
String jobTitle = rs.getString("job_title");
String createdAt = rs.getString("created_at");

// Print each row
System.out.printf("%-5d %-20s %-30s %-5d %-40s %-20s %-25s %-20s\n",
    id, fullName, email, experience, skillset, company, jobTitle, createdAt);
}

} catch (Exception e) {
    e.printStackTrace();
}

}
```

Output :-

```
Creating User using Selenium from frontend....
-----
Fetching Data to verify if user is created successfully or not ...
Mar 11, 2025 12:17:38 AM io.opentelemetry.api.GlobalOpenTelemetry maybeAutoConfigureAndSetGlobal
INFO: AutoConfiguredOpenTelemetrySdk found on classpath but automatic configuration is disabled. To enable, run your JVM with -Dotel.java.global-autoconfigure.enabled=true
ID      Full Name      Email              Exp  Skillset              Company      Job Title      Created At
-----
1      Alice Brown      alice.brown@example.com  4    Selenium, Java, Cucumber  TCS          Senior Automation Tester  2025-03-10 23:34:27
2      David Wilson      david.wilson@example.com  6    Python, Cypress, API Testing  Swiggy       Lead QA Engineer  2025-03-10 23:34:27
3      Emma Johnson      emma.johnson@example.com  2    Selenium, Jenkins, API Testing  Deloitte     Automation Tester  2025-03-10 23:34:27
4      Michael Smith      michael.smith@example.com  5    Java, Selenium, TestNG  Google       Automation Engineer  2025-03-10 23:34:27
=====
Default Suite
Total tests run: 1, Passes: 1, Failures: 0, Skips: 0
=====
```

The screenshot shows a test runner interface with a sidebar on the left listing test suites and individual tests. The main area displays the output of the 'CreateUser_Test' test, which is identical to the output shown in the previous block. The test passed successfully, and the process finished with exit code 0.

Run: DatabaseTesting

Tests passed: 1 of 1 test - 305 ms

Default Suite 305 ms

SeleniumDatabaseTesting JDBC 305 ms

DatabaseTesting 305 ms

CreateUser_Test 305 ms

Process finished with exit code 0

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